Chimère Ω: a blueprint for a physico-cognitively inspired local-first LLM runtime with entropic mode scheduling

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Abstract

We propose *Chimère* Ω, a blueprint for a local-first LLM runtime that allocates computation across four modes — fast prediction, block-diffusion refinement, symbolic verification, latent-space planning — via a scheduler driven by token-level residual entropy rather than fixed architectural choices. The blueprint integrates verified Q4 2025–Q2 2026 state-of-the-art: LeJEPA/SIGReg (Balestriero & LeCun, arXiv:2511.08544), hybrid Gated-DeltaNet + sparse-attention backbones (Qwen3-Next, Kimi-Linear), test-time memory optimisation (Titans arXiv:2501.00663 → Nested Learning / Hope arXiv:2512.24695), block-diffusion–AR fusion (BD3-LMs arXiv:2503.09573), auxiliary-loss-free MoE routing (arXiv:2408.15664), and the vLLM V1 production stack. We then propose three design hypotheses, framed explicitly as hypotheses to be tested rather than theorems: a *free-energy-under-censorship* training objective, a *renormalisation-group view of fine-grained MoE routing*, and an *A6-persistent, A3-censored memory-write policy* that ties episodic consolidation to symbolic verification. The physics inspiration from the author's earlier ECI framework (entanglement–complexity–information, v5 phenomenological and v6 formal companions) is used as analogical scaffolding, not mathematical derivation; this distinction is enforced throughout. We conclude with a 24-month phased roadmap whose first phase is immediately executable on consumer hardware (RTX 5060 Ti, 16 GB VRAM) via the author's existing `chimere-server` Rust FFI runtime and `ik_llama.cpp` backend.

1 Introduction and scope

The public debate on large-model design through late 2025 has often been framed as a dichotomy: *predict* (autoregressive next-token, exploiting pre-training redundancy) versus *compute* (test-time-compute, refinement, program synthesis). We argue this dichotomy is a false opposition. Both are projections of the same underlying quantity, the *information bits allocated per output token*, and the useful design question is how to *modulate* this allocation as a function of local residual entropy.

This paper proposes an explicit blueprint, *Chimère* Ω, for a runtime that schedules between four modes:

M1. Fast prediction ($\sim 5\,\mathrm{ms}$/token) when the per-token residual entropy $H(r_t)$ is low (e.g. < 0.5 bit): a Gated-DeltaNet dominated backbone with MoE top-K selection.

M2. Block-diffusion refinement ($\sim 30\,\mathrm{ms}$/block) when $H(r_t) \in [2, 4]$ bits: BD3-style diffusion over fixed-size blocks with preserved KV cache.

M3. Symbolic verification ($\sim 200\,\mathrm{ms}$/claim) when the candidate output is formally testable: a Scallop-lite provenance-semiring check, or Lean-style proof verification.

M4. Latent-space planning (world-model rollout) for action-conditioned tasks: V-JEPA-style predictor operating on frozen embeddings.

The contribution of this paper is not a new theorem or a new fundamental constant. It is three distinct things, clearly delineated: **(i)** a curated landscape of verified Q4 2025–Q2 2026 state-of-the-art with citations cross-checked against arXiv and publisher records (§2); **(ii)** an explicit architecture integrating those components with an entropy-driven scheduler (§3); **(iii)** three *design hypotheses* framed as falsifiable statements, inspired by the author's physics-motivated ECI framework but presented as analogical scaffolding, not mathematical derivation (§4).

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What this paper does not claim. We do not claim: a new equation of physics, a new fundamental constant, a mathematical identification between physics axioms (ECI A3 censorship, A6 persistence) and machine-learning loss terms, nor a theorem that the proposed runtime attains a particular benchmark score. The relation to ECI is motivational; the ECI v5 and v6 companion papers live at the cosmology end of the project, and a rigorous mathematical bridge to the machine-learning objectives of this paper is *not* established in either direction.

2 The verified 2025–2026 landscape

Five programmes dominate the Q4 2025–Q2 2026 state of the art and together define the engineering envelope for a local-first LLM runtime.

2.1 Self-supervised representation learning: the JEPA line

LeJEPA (Balestriero and LeCun, arXiv:2511.08544, November 2025) [1] proves that the optimal distribution on joint embedding predictor architecture (JEPA) representations is the isotropic Gaussian $\mathcal{N}(0, I_d)$ and proposes *Sketched Isotropic Gaussian Regularisation* (SIGReg) as a linear-cost surrogate:

$$\mathcal{L}_{\text{SIGReg}} = \frac{1}{M} \sum_{m=1}^{M} \mathcal{D}\Big(\hat{\varphi}_{z_m}(t),\, e^{-t^2/2}\Big), \qquad z_m = \langle u_m, s_\phi(x)\rangle,\ u_m \sim \text{Unif}(S^{d-1}) \tag{1}$$

where $\hat{\varphi}$ is the empirical characteristic function and $\mathcal{D}$ a distance to the characteristic function of $\mathcal{N}(0,1)$. By the Cramér–Wold theorem, matching all one-dimensional projections to $\mathcal{N}(0,1)$ forces isotropy of the d-dimensional distribution. This removes the heuristic stack (stop-gradient, EMA teacher, asymmetric predictor) that had characterised earlier JEPA variants.

2.2 Hybrid linear + attention backbones

The consensus architectural pattern through Q2 2026 is a three-to-one ratio of linear-attention layers (Gated DeltaNet [4], Kimi Delta Attention [5], Mamba-family SSMs) interleaved with sparse or full-attention layers. The Gated-DeltaNet update rule is an online delta-descent step,

$$S_t = \alpha_t \left(S_{t-1} - \beta_t (S_{t-1} k_t) k_t^\top\right) + \beta_t\, v_t k_t^\top \tag{2}$$

with $\alpha_t \in (0,1)$ a SiLU-gated forget factor and β_t a token-level learning rate; Kimi Delta Attention generalises α_t to a channel-wise vector $\alpha_t \in \mathbb{R}^d$ combined with a diagonal-plus-low-rank transition matrix. Mamba-3 (arXiv:2603.15569) extends the state to $h_t \in \mathbb{C}^d$ with a data-dependent rotation $A_t = \text{diag}(e^{i\theta_t} e^{-\alpha_t})$, which lifts the TC^0 expressivity bound of real-valued SSMs.

2.3 Memory: Titans → Nested Learning / Hope

Titans [2] formulates long-term memory as a test-time online optimisation with momentum and forgetting:

$$S_t = \eta_t\, S_{t-1} - \theta_t\, \nabla \ell(M_{t-1}; k_t, v_t), \quad M_t = (1-\alpha_t)\, M_{t-1} + S_t \tag{3}$$

with associative loss $\ell(M; k, v) = \|M(k) - v\|^2$. Nested Learning / Hope (Behrouz et al., NeurIPS 2025) [3] generalises this: a model is a stack of nested optimisation problems, each level ℓ minimising a loss $\mathcal{L}_\ell(\theta_\ell; C_\ell)$ where θ_ℓ parameterises the optimiser of level $\ell - 1$. The Hope reference implementation attains lower perplexity than Mamba-2 and standard transformers on long-context benchmarks and is explicitly designed with multiple-frequency update rates for continual learning.

2.4 Generation: block-diffusion, not pure diffusion

Block Diffusion (BD3-LMs, Arriola et al., ICLR 2025 Oral) [8] decomposes a sequence of length $L = B \cdot b$ into B blocks; intra-block diffusion is masked, inter-block factorisation is autoregressive,

$$\log p_\theta(x) \;\geq\; \sum_{b=1}^{B} \mathcal{L}_{\text{diffusion}}\big(x^b,\, x^{<b},\, \theta\big)\,. \tag{4}$$

The critical practical property is that BD3 preserves the KV cache across blocks, which pure discrete-diffusion cannot. The 2026 production deployments (LLaDA 2.0, Mercury 2) confirm this pattern dominates the diffusion-LM frontier.

2.5 MoE: ultra-sparse with auxiliary-loss-free balancing

The auxiliary-loss-free load balancing of Wang et al. [7], deployed in DeepSeek-V3 [6], uses an additive bias b_i that enters the top-K selection but not the gating value,

$$\tilde{s}_{i,t} = s_{i,t} + b_i, \quad g_i = \mathbb{1}[\tilde{s}_{i,t} \in \mathrm{TopK}], \quad b_i \leftarrow b_i - \gamma\,\mathrm{sign}(\mathrm{load}_i - \overline{\mathrm{load}}). \tag{5}$$

The update is applied with $\gamma = 10^{-3}$ during early pre-training and annealed to zero. Kimi K2 [9] uses 384 routed experts, the highest publicly-documented count in an open-weights model as of April 2026.

2.6 Runtime stack and quantisation

vLLM V1 unifies prefill–decode scheduling with PagedAttention at 16-token block granularity and first-class hybrid-model support (Qwen3-Next, Granite-4.0). EAGLE-3 (arXiv:2503.01840) [10] provides $3-6\times$ speculative-decoding speedup over autoregressive baselines via multi-layer feature fusion. For quantisation, SpinQuant [11] learns Hadamard rotations via Cayley parameterisation on the Stiefel manifold to close the W4A4KV4 gap to ~ 3 points vs. FP16. KIVI [12] gives 2-bit K (per-channel) / 2-bit V (per-token) KV-cache compression with $\sim 2.6\times$ memory reduction at negligible perplexity cost.

3 Chimère Ω blueprint

We now specify the blueprint, organised around three dualities and an entropy-driven scheduler.

3.1 Three dualities

D1. Predict vs. compute. A token with low residual entropy is optimally served by a fast M1 (prediction) pass; a token with high residual entropy warrants M2 (refinement) or M3 (verification). The scheduling variable is $H(r_t)$.

D2. Memorise vs. compress. Following IBLM [14], learning alternates between a memorisation phase (capacity expansion) and a compression phase (entropy reduction); we propose that *inference* should likewise alternate via an episodic write/consolidate policy.

D3. Local vs. global. Fine-grained MoE routing is a coarse-graining step: as depth increases, low-level details are integrated out toward semantic features. We suggest (Hypothesis 2) that this be formalised as a discrete-layer renormalisation-group flow.

3.2 Architecture

A 96-layer backbone interleaving GDN/KDA layers (75%) with sparse full-attention (15%) and dense layers (10%), fine-grained MoE (512 routed experts + 2 shared + 4 proposed *censor* experts) with auxiliary-loss-free balancing, multi-head output (MTP 4 for AR generation, BD3 for refinement, JEPA predictor for planning, verification head for symbolic checks), and a 4-tier memory hierarchy (L1 KV cache, L2 Titans LMM, L3 FHRR hash table, L4 EM-LLM episodic segmentation).

Censor experts. Four MoE experts are reserved for tokens with high residual entropy. Their selection score adds a residual-entropy term:

$$s_{i,t} = \mathrm{sigmoid}(u_t^\top e_i) + b_i + \mathbb{1}_{\{i \in \mathrm{censor}\}} \cdot \kappa \cdot H(r_t). \tag{6}$$

This is a design choice, not a theorem: it is engineered to concentrate out-of-distribution tokens into experts whose weights can carry additional regularisation (spectral-norm constraints, narrower weight initialisation). Whether this improves out-of-distribution behaviour in practice is a testable prediction.

3.3 Entropy-driven scheduler

The runtime computes $H(r_t) = -\sum_v p_\theta(v|\cdot) \log p_\theta(v|\cdot)$ (or a top-k approximation) per token and dispatches:

$H(r_t)$ range	Mode(s) activated	Wall-clock target
< 0.5 bit	M1 only	5 ms/token
$[0.5, 2)$ bits	M1 + EAGLE-3 speculative	10 ms/token
$[2, 4)$ bits	M2 (BD3 block refinement)	30 ms/block
≥ 4 bits	M2 + M3 + L3 retrieval	200+ ms/claim

The thresholds are engineering proposals to be calibrated empirically; the design principle is that *marginal expected information gain of adding mode $N+1$ must exceed its wall-clock cost.* This is implementable today with the `chimere-server` runtime FFI-ed to `ik_llama.cpp`; no new training is required to deploy the M1/M2 split.

4 Three design hypotheses

The following three expressions are framed as *hypotheses*, not theorems. Each inherits motivation from the author's ECI framework axioms (A3 censorship-like suppression; A6 persistence-like consolidation); the mapping from cosmological axioms to machine-learning objectives is *analogical* and is not derived. Each hypothesis yields a concrete experimental programme with a binary outcome.

4.1 H1 — Free energy under censorship (FEC)

Hypothesis 1 (Free energy under censorship, FEC)**.** *Training a Chimère Ω model with the modified objective*

$$\mathcal{L}_{\text{FEC}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x)]}_{\text{NTP}} + \beta\, I_{\text{SAE}}(Z_{\text{free}}; X) - \gamma\, \langle V(M) \rangle + \lambda\, \mathcal{L}_{\text{SIGReg}}(Z) \tag{7}$$

where I_{SAE} is the mutual information between the subset of SAE features not causally related to the prediction target and the input, V is a learned scalar functional on memory states M, and $\mathcal{L}_{\text{SIGReg}}$ is Eq. (1)*, yields equal or lower bits-per-character on out-of-distribution evaluation sets at matched FLOPs versus the NTP-only baseline.*

Motivation 1. The ECI framework suggests "censoring" information that cannot be predicted, via an activation function Θ in the cosmological horizon setting. The analogue proposed here is to penalise SAE features that carry no prediction signal, weighted by β. The memory-potential term $\gamma\langle V(M)\rangle$ is the ML analogue of the persistence axiom A6 in ECI, rewarding long-lived informative memory content. The SIGReg term closes the isotropy condition on latent representations.

Falsification. H1 is falsified if, after hyperparameter tuning of (β, γ, λ), the FEC objective fails to beat matched-FLOPs NTP on at least one of: BPC on a held-out domain, Arc-AGI-2 single-sample accuracy, or RULER at 128k context. A negative result would not invalidate the ECI framework (which lives in a different domain) but would falsify the design analogy.

4.2 H2 — MoE routing as a discrete renormalisation-group flow

Hypothesis 2 (RG-flow MoE)**.** *Let $b_i^{(\ell)}$ denote the load-balancing bias of expert i at layer ℓ. If the interlayer update is augmented by a coarse-graining operator $\mathcal{R}[\cdot]$ that maps fine-grained biases at layer $\ell - 1$ to effective biases at layer ℓ,*

$$\frac{db_i^{(\ell)}}{d\ell} = -\gamma\, \text{sign}\Big(\text{load}_i^{(\ell)} - \overline{\text{load}}^{(\ell)}\Big) + \mu\, \mathcal{R}\Big[b_i^{(\ell-1)}\Big], \tag{8}$$

then training stability improves and the emergent expert specialisation follows a layer-dependent power law $\kappa^{(\ell)} \propto \ell^{-\alpha}$.

Motivation 2. Fine-grained MoE segmentation plus auxiliary-loss-free balancing is already a form of layer-wise reorganisation. Interpreting it as a Kadanoff-like block-spin map suggests that the bias update should carry information between layers rather than be reset per layer.

Falsification. H2 is falsified if, on a matched-FLOPs pretraining run, the added $\mathcal{R}$ term neither reduces expert-imbalance variance nor produces a consistent power-law in $\kappa^{(\ell)}$.

4.3 H3 — Engram write policy with A6-persistence and A3-censorship

Hypothesis 3 (Engram write policy)**.** *An episodic-memory write policy of the form*

$$\pi_{\text{write}}(e_t \mid h_t) = \sigma\big(w_0 + w_1 \,\text{Surprise}_B(h_t) + w_2 \, V_{\text{verify}}(h_t) - w_3 \, \mathcal{C}_{\text{A3}}(h_t)\big) \tag{9}$$

with Surprise_B *the EM-LLM bayesian surprise,* V_{verify} *a symbolic-check score, and* $\mathcal{C}_{\text{A3}}$ *a hallucination-risk score from a trained SAE, outperforms a threshold-only write policy on long-horizon factual consistency benchmarks.*

Motivation 3. A memory that writes every surprising event risks consolidating hallucinations; a memory that writes nothing but verified claims misses creative long-horizon coherence. H3 proposes a trainable trade-off with four factors, two of which (V_{verify}, $\mathcal{C}_{\text{A3}}$) are concrete proxies for the ECI A3 censorship axiom, and one of which (Surprise_B) is a direct instantiation of the A6 persistence condition restricted to surprising events.

Falsification. H3 is falsified if an ablation that removes either V_{verify} or $\mathcal{C}_{\text{A3}}$ yields equal benchmark performance.

4.4 Honest note on ECI as scaffolding

All three hypotheses use the ECI v3 axioms (A3, A6) as *linguistic and architectural scaffolding.* The rigorous ECI v5 (phenomenological) and v6 (formal) companion papers inhabit a wholly different scientific domain (cosmology and type-II crossed-product algebras). An explicit attempt at a mathematical bridge from the ECI Jacobson–EGJ internal production ($d_i S$) to a machine-learning source term failed under a dedicated audit (see [15], registered in the author's `FAILED.md` as entry F-14). The ML hypotheses in this paper therefore rely on the *form* of the ECI axioms, not on any derivation from them. This distinction is load-bearing and must not be blurred.

5 Roadmap and feasibility on consumer hardware

5.1 Hardware budget (single-author context)

The author's workstation (i5-14600KF, 32 GB DDR4, RTX 5060 Ti 16 GB VRAM) and the existing `chimere-server` Rust FFI to `ik_llama.cpp` define the hardware envelope. For a $\sim$ 30B-active-parameter upcycled hybrid model at IQ3_S weights with Q8_0 on routers and final LayerNorm, the total memory footprint is $\sim$ 34–38 GB, which fits with aggressive DDR-offload of cold experts; expected throughput $\sim$ 8–15 tok/s. From-scratch pretraining is infeasible at this budget; continue-training via Drop-Upcycling [13] from a published base model (e.g. Qwen3-Next-80B-A3B) on Vast.ai spot instances remains feasible at $\sim$ \$50k–\$200k.

5.2 Four-phase plan

Phase 1 (months 0–3). Runtime integration: entropic scheduler M1/M2 on the existing `chimere-server` stack. No new training. Gold-path demonstration on a Qwen3-Next-80B upcycle.

Phase 2 (months 3–8). Drop-Upcycling [13] of Qwen3-Next, addition of BD3 and JEPA predictor heads via LoRA, Engram L1–L3 integration.

Phase 3 (months 8–14). Mode M3 symbolic verification, SAE-based interpretability for censorship scores, calibrated hypothesis-H3 ablations.

Phase 4 (months 14–24). A 3–8B from-scratch training with the full FEC objective (Eq. 7) on $\sim$ 200–500B tokens via Vast.ai; this is the first stage where hypothesis H1 becomes directly testable.

5.3 What the pure assemblage already delivers

Phase 1 alone (entropic scheduler on existing components) is a strict improvement over a monolithic AR or monolithic diffusion baseline on workloads with mixed residual entropy: gains of $\sim 3-6\times$ wall-clock on typical mixed-entropy workloads follow mechanically from the EAGLE-3 + M2-selective-refinement combination. The proposed design hypotheses (H1–H3) should only be introduced after the assemblage baseline saturates, which we estimate at the end of Phase 3.

6 Limitations and honest caveats

The ECI scaffolding is analogical, not derivational. The ECI v3 axioms A3 ("cryptographic censorship") and A6 ("persistent-homology complexity") live in a cosmological setting with specific mathematical content; we use their *names and structural role* as scaffolding for the machine-learning hypotheses. An explicit attempt to derive the form of the v6 source term from the Jacobson–EGJ internal production $d_i S$ failed in a dedicated audit [15], with four sub-obstructions catalogued (categorical mismatch between state-functional complexity and extrinsic horizon geometry; Gaussian-coherent-state erasure of $|\alpha|^2$ in spread complexity; the $\delta Q = T\delta S$ Wald-entropy tautology; absence of coupling-coefficient prediction in type-II crossed-product frameworks). The present paper inherits the diagnostic weight of this audit: the ECI language is suggestive, not load-bearing.

Production-first, research-second. No hypothesis is promoted unless Phase 3 completes; Phase 1 delivers value whether or not the hypotheses hold.

Falsification is explicit. Each hypothesis in Section 4 carries a *falsification clause*; a clean failure is a valid result. The `FAILED.md` discipline inherited from the ECI project applies.

We do not claim SOTA. The blueprint targets a local-first, interpretable, and honestly scoped runtime, not a frontier-model ranking on a fixed benchmark suite. If a frontier-cloud runtime (DeepSeek, Kimi, Gemini) scores higher on any given benchmark, that is orthogonal to the value proposition here.

7 Conclusions

The frontier between *prediction* and *computation* is not a dichotomy; it is a budget. Chimère Ω proposes a runtime that allocates that budget per-token as a function of residual entropy, integrating verified 2025–2026 components (LeJEPA/SIGReg, hybrid GDN + sparse-attention backbones, Titans / Nested Learning memory, BD3 block-diffusion, auxiliary-loss-free MoE, vLLM V1 + EAGLE-3 runtime) with three design hypotheses inspired by the author's ECI framework but explicitly presented as hypotheses to be tested. Phase 1 of the roadmap is immediately executable on consumer hardware and delivers value independently of whether hypotheses H1–H3 hold; Phases 2–4 progressively introduce the hypothesis testing.

The intellectually honest deliverable of this paper is not a new theorem. It is a *plan of record* for a line of work whose first executable product uses existing verified components and whose speculative content is explicitly isolated, testable, and falsifiable. Further progress on Chimère Ω will be registered under the same discipline that produced the ECI v6.0.3 companion paper: pre-registered falsifiers, explicit failure logs, and cross-model audits before any paper update.

Acknowledgement. This blueprint was substantially drafted with the assistance of large-language-model chat and agent interfaces, with citation verification performed by WebSearch / WebFetch against arXiv and publisher records. All numerical claims of recent papers cited here were cross-checked against the primary source. No citation was retained without primary-source verification. The three design hypotheses are the author's, not the assistants'.

Data availability. This is a blueprint paper; no datasets or training runs are reported. The reference chimere-server runtime is available at `https://github.com/AIdevsmartdata/chimere`. All cited external models, datasets, and benchmarks are public and accessible from their respective arXiv / HuggingFace releases.

Conflicts of interest. The author declares no competing financial or non-financial interests.

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